Asbestos application implementation guidance

Public references checked October 15 2026. Suggested architecture, not prebuilt ORTAM code. Candidates may choose equivalent tools and should explain their choices. No private repository, production credential or production deployment is needed.

One report revision for preview and downloads

Suggested flow: employee form -> saved structured project -> uploaded lab PDFs -> extracted text or pages -> AI structured findings -> validated sample matching -> employee review -> report revision -> DOCX -> PDF conversion -> embedded PDF preview and both downloads.

Store immutable source documents and source references separately from employee corrections. Each accepted finding retains the project, laboratory report, PDF page, sample ID and layer. Keep field quantities on material zones, not lab rows. Unknown and conflicting data stay visible.

Use one validated project snapshot to build each report revision. Return the DOCX and converted PDF under the same revision. The preview displays that exact PDF. An employee edit marks the preview outdated; regeneration must complete before displaying or downloading the current revision. Show conversion failures clearly instead of substituting a stale file.

A full browser-based Word editor is outside this test. The employee edits form fields and report narrative in the app, regenerates the report, and can continue editing the downloaded DOCX in Word. Allow exporting a visibly labeled draft with unresolved exclusions; do not silently approve it.

AI extraction and deterministic checks

Ask the model for structured records with sample ID, layer, result label, reported type, reported percentage, source page and supporting source text. Validate allowed labels and data types. The model does not calculate the final estimate or decide regulatory thresholds. The app computes the supplied scenario using accepted quantities and fixed rates.

Match within a project by sample ID; retain layer-level results and all samples per material zone. Employee review resolves unmatched or contradictory records. Reimporting the same PDF must not duplicate findings or charges. Demonstrate actual model use and at least one manual correction.

For a compact assessment, a local web app and local persistent storage are acceptable. If a document converter is used, a local container service is acceptable. Document the model access and converter setup; never require the reviewer to obtain access to ORTAM systems.

Public building blocks

These are implementation references inspected through their documentation. They have not been integrated or benchmarked together for this test. Pin chosen versions, retain required licenses, and explain any dependencies.

PDF viewing and sample placement

Mozilla PDF.js parses and renders PDF pages in a browser. Use it to display input lab pages, floor plans and the generated PDF report. Its distributed package is pdfjs-dist.

Mozilla PDF.js repository

Store floor plan markers independently of the PDF: plan file ID, page number, normalized X and Y from top left, and sample ID. Recompute screen position from the displayed page dimensions. Test zoom, resize and page rotation. Resolve pin status from the accepted lab records, including positive layers and unresolved results.

pdf-lib can add text and shapes to existing PDFs for a marked plan export. It is a PDF editing library, not a Word renderer or a PDF-to-CAD converter. Use the same stored marker coordinates when drawing the exported annotations.

pdf-lib repository

Word generation

The docx library generates editable DOCX documents in JavaScript or TypeScript, including text, tables and images. Alternatively, Docxtemplater fills tagged Word templates. The supplied master contains human-readable placeholders; a candidate may convert those to tool-specific tags or reproduce its structure. Docxtemplater has an open-source core and optional modules; image features may require a module. Neither library should be assumed to provide a Word-to-PDF renderer.

docx repository

Docxtemplater repository

Docxtemplater module documentation

Word to PDF conversion

Gotenberg supplies a container-based API that converts Office documents to PDF using LibreOffice. Generate DOCX first, convert it, and use the resulting PDF for preview and PDF download. Run the converter as a separate service when the app host cannot run office software. Keep fonts consistent and inspect page breaks, tables and any plan appendix in both formats.

Gotenberg repository

Gotenberg Office conversion documentation

CAD research beyond the mapping bonus

The mapping requirement is to annotate the supplied PDF plan and connect samples to results. It does not require reconstructing CAD geometry. PDF viewing with a marker overlay is the suggested test approach. Investigating editable CAD can be proposed as later work without changing the core or bonus scoring.

CADCraft and AI agent access

CADCraft documents DXF and DWG support, drawing commands, an MCP server, inspection tools and PDF plotting or export. This gives agents a documented interface for inspecting and editing supported CAD drawings. Its README describes an early and actively developing project. A documented end-to-end PDF-to-editable-CAD import path was not established in this review; PDF export does not establish PDF import capability.

CADCraft repository and agent documentation

A possible conversion experiment

PDF-Importer-LibreCAD documents PDF vector drawing conversion to DXF. Treat it as a research candidate, not a qualified dependency: it was not installed or tested on these plans. A proposed experiment is vector PDF -> conversion -> DXF -> CADCraft inspection and edits -> exported PDF. Verify every step before claiming the pipeline works. Review dependency licenses as well as the top-level repository license.

PDF-Importer-LibreCAD repository

ezdxf is a Python library for creating and modifying DXF drawings. It is useful after geometry is available; it does not itself establish an automatic PDF-to-CAD conversion workflow.

ezdxf repository

Conversion limits

Vector PDFs and scanned image PDFs need different treatment. Recovering PDF linework does not guarantee original CAD objects, layers, dimensions or precision. Autodesk documents information and precision loss in drawing PDFs, and notes that image-based content is not converted into vector geometry by ordinary PDF import. A scan may remain an image underlay or require a separate tracing workflow. These exercise plans are illustrative and not to scale, so they cannot validate measured quantities or dimensional accuracy.

Autodesk explanation of PDF import limitations

If explored later, verify page scale, units, rotation, known reference lengths, text and line fidelity, and opening and saving the resulting drawing. AI can operate exposed tools, but a tool connection does not prove the conversion preserved the source.